

Wanhao Liu

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RESEARCH INTERESTS

Vision-Language-Action Models, World Action Models, Embodied Intelligence, Visuomotor Control, Surgical Robotics, and Medical Robot Autonomy.

EDUCATION

Guangdong University of Technology (GDUT)

Sep 2023-Jul 2027

B.Eng. in Electronic Science and Technology

GPA: **88.20/100** | Rank: **9/221**

PUBLICATIONS

JOURNAL

- [1] Luo, L., **Liu, W.**, et al. "Prescribed-time fault-tolerant attitude control for tiltrotor UAV with input saturation and mismatched disturbances." **Control Engineering Practice, SCI-Q2, IF=5.3 (Accept)**
- [2] Zhou, R*, **Liu, W***, et al. "Surg-UniWorld: A Unified Surgical World Model with Multimodal Control Experts." **IEEE Transactions on Medical Imaging. SCI-Q1, Top, IF=12.4 (under review)**
- [3] Zhou, R*, Song, Q*, Ma, Y*, Wang, A., **Liu, W.**, et al. "SurgLAT: Surgical Latent Attention Tracking for Depth-Aware Robotic Laparoscope Control." **IEEE Transactions on Medical Imaging. SCI-Q1, Top, IF=12.4 (under review)**

CONFERENCE

- [4] **Liu, W.***, et al. "CrossScope: A Role-Asymmetric World Model for Joint Dual-Scope Surgical Video Prediction." AAAI 2027, under review. **(CCF-A)(under review)**
- [5] **Liu, W.***, et al. "AC-MASAC: An attentive curriculum learning framework for heterogeneous UAV swarm coordination." AAMAS, Paphos, Cyprus. **(CCF B Oral, Flagship Conference on Multi-Agent Systems)(Accept)**
- [6] Pan, W*, **Liu, W.***, et al. "NCGR: Noise-Conditional Gated Rectification for Camera Extrinsic Perturbations in BEV 3D Object Detection." AAAI 2027, under review. **(CCF-A)(under review)**
- [7] Lin, J*, Pan, Z*, **Liu, W.***, et al. "EndoWAM: A Grounded World-Action Model for Generalizable Endoscopic Navigation." AAAI 2027, under review. **(CCF-A)(under review)**
- [8] Dai, J., Cui, Y., **Liu, W.**, et al.. "Conflict prioritized-based experience replay soft actor-critic algorithm for unsignalized intersections coordination." CVCI 2025, Qingdao. **(Accept)**
- [9] Zhang, Y., **Liu, W.**, & Liu, L. (2024). "Improved heuristic JPS algorithm for path planning in intelligent warehouse robot." ICEMCE 2024, Xi'an. **(Accept)**

PATENTS

- [10] Li, P., **Liu, W.**, Yin, S., Tian, X., Dai, J., & Lu, R. Path planning method and system for heterogeneous UAV swarms. Chinese patent application, CN121835804A. Filed on November 6, 2025.
- [11] Zhang, Y., Liu, L., **Liu, W.**, Luo, T., & Yin, S. Shopping mall store recommendation method, apparatus, device, and medium. Chinese patent application, CN119513421A. Filed on November 20, 2024.

RESEARCH EXPERIENCE

Executable World-Action Modeling for Robot-Assisted Endoscopic Intervention [\[Webpage1\]\[2\]](#)

Research Assistant | Advised by [Prof. Hongliang Ren](#), CUHK Jan 2026-Present

- Formulated generalizable robotic endoscopic navigation as an executable **World-Action Model (WAM)** that jointly predicts future **visual dynamics**, **target-region evolution**, and 3-DoF action chunks
- Designed a future-grounded **Video DiT** with discrete action and target-grounding experts in one denoising pass; integrated **role-asymmetric Mother-Child routing** to transfer geometry-guided motion and pose-aligned appearance for **dual-scope ERCP** prediction
- Constructed EndoMotion and paired dual-scope ERCP benchmarks with 515 teleoperated trajectories (**0.98M frames**; **7.84M viewpoint-augmented**) and 352 synchronized episodes; achieved **80.2% closed-loop navigation success** and **33.0% lower endpoint-motion error**

Surg-UniWorld: Unified Surgical World Modeling with Multimodal Control Experts [\[Webpage\]](#)

Research Assistant | Advised by [Prof. Hongliang Ren](#), CUHK Jan 2026-Present

- Built a controllable laparoscopic **world model** conditioned on text, first-frame appearance, hierarchical masks, and flexible **edge/depth/flow** controls
- Designed **Surg-ARCA** with region-aware anchors, modality-specific experts, and stage-wise **Wan2.2 DiT** injection for anatomically and temporally consistent instrument-tissue interactions
- Curated Cholec80-SurgWAM: **6,001 clips**, **294K frames**, and **574K hierarchical masks** with multimodal controls; **reduced FVD/FID by 11.0%/34.9%** over the strongest baseline

Attentive Curriculum Learning Framework for Heterogeneous UAV Swarm Coordination [\[Code\]](#)

Research Assistant | Advised by [Prof. Panshuo Li](#), GDUT Mar 2025-Oct 2025

- Proposed a **Multi-Agent Reinforcement Learning (MARL)** framework for heterogeneous UAV swarm coordination to address asymmetric agent dependencies and unstable policy convergence
- Designed and implemented **AC-MASAC**, a novel MARL framework with **role-aware attention** and **curriculum learning** to mitigate sparse rewards and catastrophic forgetting
- Outperformed **MASAC**, **MADDPG**, and **RRT*** in SR, FKR, and SMT within a custom 700 × 600 m simulation, improving **FKR by over 15%** and **final average reward by ~ 82%** in ablation studies

SCIENTIFIC COMPETITION

National University Students Smart Car Competition Jun 2025-Sep 2025

Project Leader | **National Second Prize (Top5% out of 200+teams)** [\[Code\]](#)

- Used CNN and PP-YOLOE models for lane line and target detection, which supported robotic arm kinematic modeling and trajectory planning
- Utilized a hierarchical control framework with a high-level planner for dual infrared sensor fusion and task decision-making, and a low-level controller for accurate trajectory tracking during multi-task execution
- Deployed the system on the Jetson Orin Nano platform with a coprocessor and ROS-based real-time control, achieving remarkable stability and task completion performance in the final testing

AWARDS & HONORS

First-Class Scholarship (Top 3%) , Guangdong University of Technology	Sep 2024-Sep 2025
Third Prize, National University Student Smart Car Competition, South China Division	Jul 2024-Aug 2024
Advanced Individual Award, Guangdong University of Technology	Sep 2023-Sep 2024

TECHNICAL SKILLS

Python, C++, LaTeX, MATLAB, Linux | Tensorflow, NumPy, Git, PyTorch, Anaconda, ROS1/ROS2 (proficient)